Supplementary Materials

Frailty screening in dermato-oncology practice: a modified Delphi study and a systematic review of the literature.

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Table S1. Systematic literature search to identify frailty screening tools

Database searched: PubMed, Embase (Excerpta Medica Database), Cochrane Library, SUMsearch, PsycINFO and Trip Database
Period of time searched: Articles published up to 16-5-2019
Limit(s): none.
Search details (search strategy shown for PubMed, similar search terms were used for other databases): *("frailty"[MeSH Terms] OR "frailty"[All Fields]) AND ("diagnosis"[Subheading] OR "diagnosis"[All Fields] OR "screening"[All Fields] OR "mass screening"[MeSH Terms] OR ("mass"[All Fields] AND "screening"[All Fields]) OR "mass screening"[All Fields] OR "screening"[All Fields] OR "early detection of cancer"[MeSH Terms] OR ("early"[All Fields] AND "detection"[All Fields] AND "cancer"[All Fields]) OR "early detection of cancer"[All Fields]) AND ("neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "cancer"[All Fields])*

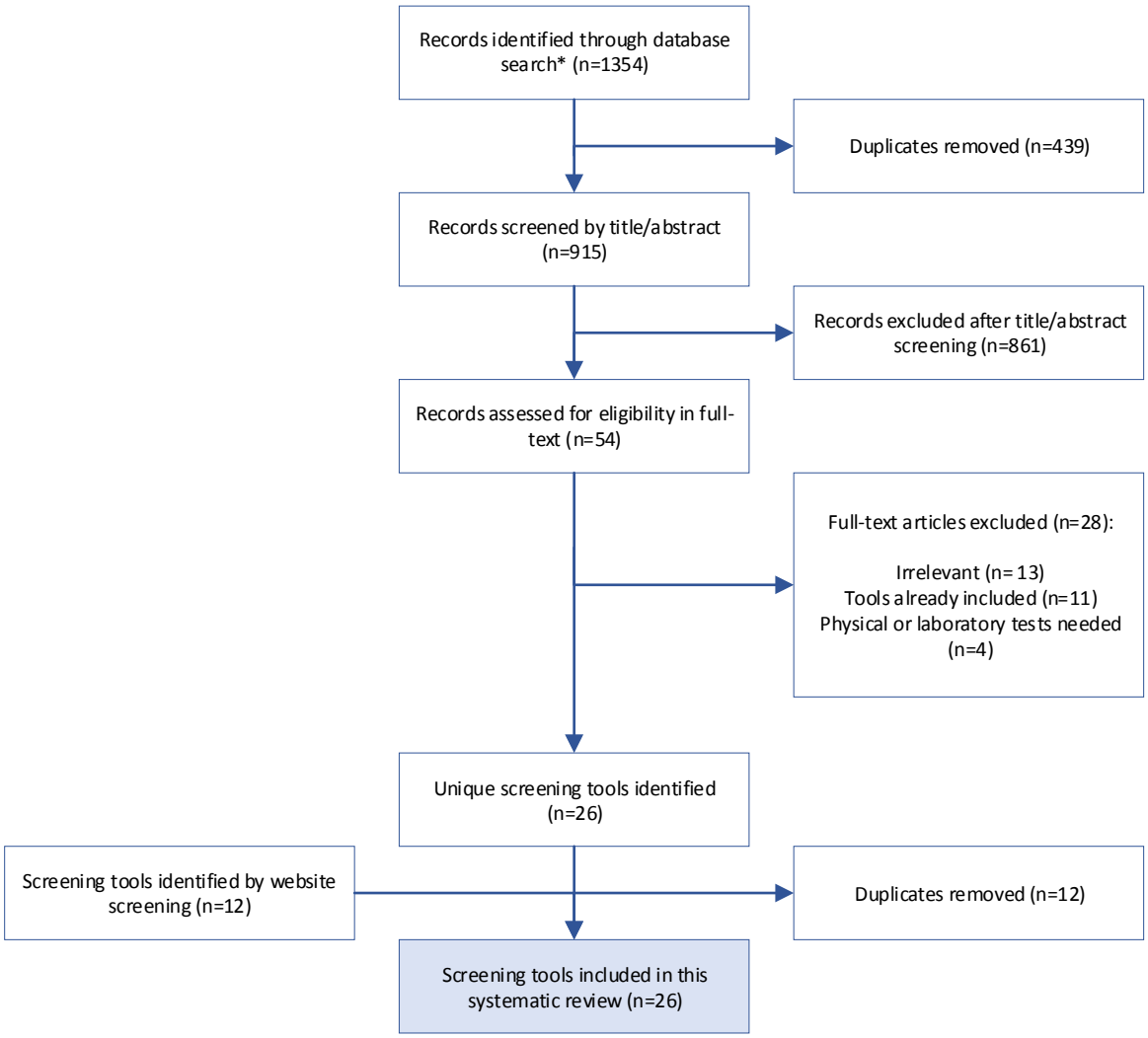
Table S2. List of relevant websites and organisations included in the systematic search to identify frailty screening tools

Organisation	Country/region	URL
International society of geriatric oncology (Société Internationale d'Oncologie Gériatrique; SIOG)	Worldwide	http://www.siog.org/content/comprehensive-geriatric-assessment-cga-older-patient-cancer
World Health Organisation (WHO)	Worldwide	http://www.who.int/cancer/en http://www.who.int/ageing/en
European Organisation for Research and Treatment of Cancer (EORTC)	Europe	www.eortc.org
Organisation of European Cancer Institutes (OECI)	Europe	http://oeci.eu/AboutOeci.aspx
European CanCer Organisation (ECCO)	Europe	http://www.ecco-org.eu
European Society of Medical Oncology (ESMO)	Europe	http://www.esmo.org
European Society for Radiotherapy & Oncology (ESTRO)	Europe	https://www.estro.org
European Union Geriatric Medicine Society (EUGMS)	Europe	http://www.eugms.org
European Union (EU), including European Commission (EC) and the European Innovation Partnership on Active Health and Ageing (EIP on AHA)	Europe	https://ec.europa.eu/eip/ageing/library/guidelines-instruments-screening-and-diagnosis-frailty-and-functional-decline
National Cancer Institute (NCI)	United States	https://www.cancer.gov
National Comprehensive Cancer Network (NCCN)	United States	https://www.nccn.org/professionals/physician_gls/default.aspx#senior
American Cancer Society (ACS)	United States	https://www.cancer.org
American Society of Clinical Oncology (ASCO)	United States	https://www.asco.org/practice-guidelines/cancer-care-initiatives/geriatric-oncology
American Geriatrics Society (AGS)	United States	https://www.americangeriatrics.org
Netherlands Comprehensive Cancer Organisation (Integraal Kankercentrum Nederland; IKNL)	The Netherlands	www.iknl.nl
Dutch Cancer Society (KWF)	The Netherlands	https://www.kwf.nl
Dutch Geriatrics Society (Nederlandse Vereniging voor Klinische Geriatrie; NVKG)	The Netherlands	https://www.nvkg.nl
Geriatric Oncology Working Group (British Geriatrics Society)	Great Britain	http://www.bgs.org.uk/oncogeriatrics/sections/specialinterest/oncogeriatrics
British Geriatrics Society	Great Britain	http://www.bgs.org.uk

Table S3. Statements as presented in round 2 of the modified Delphi procedure on the requirements a frailty screening tool should fulfil in older dermato-oncology patients

The screening tool should have appropriate measurement properties* **following the COSMIN checklist, which included: internal consistency, reliability, measurement error, content validity, structural validity, hypothesis testing, cross-cultural validity, criterion validity, responsiveness, and interpretability*
The screening tool should be available in multiple languages
The screening tool should include items related to comorbidities
The screening tool should include items related to medication use or polypharmacy
The screening tool should include items related to mobility and falls
The screening tool should include items related to (instrumental) activities of daily living
The screening tool should include items related to neurosensory deficits
The screening tool should include items related to cognitive function (e.g. dementia and/or delirium risk)
The screening tool should include items related to mood, anxiety and/or depression
The screening tool should include items related to nutritional status
The screening tool should include items related to social support system (i.e. relatives/friends) or health care related support system (community nursing)
The screening tool should include items related to the ethnic or cultural background
The screening tool should be filled in by a physician/physician assistant/nurse (practitioner)
The screening tool should be filled in by a patient or relative/friend
The screening tool items should be easy to understand for the person filling in the screening tool; e.g. also suitable for use in a low-educated and/or low-literature population
The screening tool should be quickly performed in daily dermatologic practice
The outcome of the screening tool should be easy to interpret
The outcome of the screening tool should have a relevant impact in daily dermato-oncologic practice (clinical relevance)
The outcome of the screening tool should have a predictive value concerning life expectancy

Figure S1. Flow-chart of the systematic search to identify frailty screening tools



* PubMed (n=418), Embase (n=290), Cochrane Library (n=30), SUMsearch (n=305), PsycINFO (n=7) and Trip Database (n=304).

Table S4. Sensitivity and specificity of the included frailty screening tools

Frailty Screening Tool	Study	n	Sensitivity	Specificity	Cut-off point CGA	Components CGA
G8	Bellera, 2012 [18]	364	85	65	≥1	ADL, IADL, MMSE, GDS-15, MNA, CIRS-G, and GUG
	Bellera, 2012 [18]	364	92	45	≥2	ADL, IADL, MMSE, GDS-15, MNA, CIRS-G, and GUG
	Luce, 2012 [59]	211	65	3	≥1	ADL, IADL, MMSE, GDS, MNA, CCI, social evaluation, biochemical lab, Mob-T, WHOQOL, pain, polypharmacy
	Pottel, 2012 [72]	51	86	75	≥2	ADL, IADL, MMSE, GDS, MNA, CIRS-G and Tinetti balance and gait
	Baitar, 2013 [12]	170	92	52	≥1	ADL, IADL, MMSE, GDS-30, MNA, CIRS-G, TUG, and MOS-SSS
	Hamaker, 2014 [49]	108	69	79	≥2	ADL (Barthel), IADL, MMSE, GDS-30, TUG, polypharmacy, BMI/weight loss, and F-Sozu
	Holmes, 2014 [51]	50	70	71	≥2	ADL, IADL, CLOX 1, CLOX 2, TMTA, TMTB, HADS-anxiety and depression, MNA, HCT-CI, polypharmacy, SPPB, and MOS
	Kenis, 2014 [8]	937	87	59	≥2	ADL, IADL, MMSE, GDS-15, MNA-short form, and CCI
	Liuu, 2014 [71]	518	87	60	≥2	ADL, IADL, MMSE, mini-GDS, MNA, CIRS-G, and GUG
	Soubeyran, 2014 [9]	1597	77	64	≥1*	ADL, IADL, MMSE, GDS-15, MNA, CIRS-G, and TUG
	Velghe, 2014 [56]	50	89	100	≥1	ADL, IADL, MMSE, GDS-4, MNA, and any falls in the previous year
	Smets, 2014 [58]	398	78	68	≥1	ADL (Barthel), IADL, MMSE, GDS-15, food intake, weight loss, medication
	Dubruille, 2015 [10]	90	80	56	≥2	ADL, IADL, MMSE, GDS-4, MNA, CCI, TUG, Mob-t, HADS, and polypharmacy
	Kenig, 2015 [73]	135	97	44	≥2	ADL, IADL,BOMC, GDS, MNA, CCI, TUG, and CDT
	Martinez-Tapia, 2016 [19]	729	87	58	≥2	ADL, IADL, MMSE, Mini-GDS, MNA, CIRS-G, and TUG

	Hentschel, 2016 [50]	84	38	63	≥2	IADL, cognition, NCCN Distress Thermometer, MNA-short form, polypharmacy, and falls
	de Thezy, 2017 [70]	49	100	73	≥1	ADL, IADL, MMSE, GDA, TUG, and malnutrition
	Osborne, 2017 [54]	178	44	84	≥1	ADL, IADL, MNA, BMI, CCI, polypharmacy, World Helath Organization Performance status, and medication
	Pamoukdjian, 2017 [60]	269	90	35	≥1	ADL, simplified IADL, MMSE, mini-GDS, BMI, albumin, and SPPB
	Pamoukdjian, 2017 [60]	269	93	29	≥2	ADL, simplified IADL, MMSE, mini-GDS, BMI, albumin, and SPPB
	van Loon, 2017 [61]	123	92	26	≥2	ADL, IADL, MMSE, GDS, MNA, CIRS-G, TUG, ECR, CDT, and fluency
	Russo, 2018 [11]	282	89	50	≥3	ADL (Barthel), IADL, MMSE, GDS-15, MNA, CIRS-G, TUG, CDT, Tinetti Scale, Gijon Scale and risk of falls
	Yokom, 2018 [57]	28	73	80	≥2	IADL, cognition, mood, nutrition, CCI, social vulnerability, polypharmacy, mobility and falls
Modified G8	Martinez-Tapia, 2016 [19]	729	89	79	≥2	ADL, IADL, MMSE, Mini-GDS, MNA, CIRS-G, and TUG
	Pamoukdjian, 2017 [60]	269	89	65	≥1	ADL, simplified IADL, MMSE, mini-GDS, BMI, albumin, and SPPB
	Pamoukdjian, 2017 [60]	269	92	36	≥2	ADL, simplified IADL, MMSE, mini-GDS, BMI, albumin, and SPPB
GFI	Kellen, 2010 [48]	113	39	86	≥2	ADL (Barthel), IADL, MMSE, and GDS
	Baitar, 2013 [12]	170	66	87	≥1	ADL, IADL, MMSE, GDS-30, MNA, CIRS-G, TUG, and MOS-SSS
	Smets, 2014 [58]	398	76	73	≥1	ADL (Barthel), IADL, MMSE, GDS-15, food intake, weight loss, and medication
	Hamaker, 2014 [64]	73	69	76	≥1	IADL, MMSE, GDS-15, BMI, CCI, and polypharmacy
	Kenig, 2015 [53]	135	64	86	≥2	ADL, IADL,BOMC, GDS, MNA, CCI, TUG, and CDT

	van Loon, 2017 [61]	123	74	52	≥2	ADL, IADL, MMSE, GDS, MNA, CIRS-G, TUG, ECR, CDT, and fluency
SAOP2	Extermann, 2009 [20]	31	100	40	NR	ADL, IADL, MMS, GDS, CES-D, MNA, CIRS-G, self-rated health, ECOG PS
	Russo, 2018 [11]	282	94	50	≥3	ADL (Barthel), IADL, MMSE, GDS-15, MNA, CIRS-G, TUG, CDT, Tinetti Scale, Gijon Scale and risk of falls

Abbreviations: ADL, Activities of Daily Living; BMI, Body Mass Index; BOMC, Blessed Orientation-Memory-Concentration; CCI, Charlson Comorbidity Index; CDT, Clock Drawing Test; CES-D, Center for Epidemiologic Studies Depression Scale; CGA, comprehensive geriatric assessment; CIRS-G, Cumulative Illness Rating Scale for Geriatrics; CLOX, Clock Draw Test; ECOG PS, Eastern Cooperative Oncology Group Performance Scale; ECR, Enhanced Cued Recall; F-Sozu, Fragebogen zur Erfassung der sozialen Unterstützung; G8, Geriatric-8; GDS, Geriatric Depression Scale (followed by the number of items); GFI, Groningen Frailty Index; GUG, Get Up and Go; HADS, Hospital Anxiety and Depression Scale; HCT-CI, Hematopoietic and Cellular Therapy-Comorbidity Index; IADL, Instrumental Activities of Daily Living; mG8, modified Geriatric-8; Mini-GDS, Mini Geriatric Depression Scale; MMSE, Mini-Mental Status Exam; MNA, Mini Nutritional Assessment; Mob-T, Mobility-tiredness scale; MOS, Medical Outcomes Survey; MOS-SSS, Medical Outcomes Study Social Support Scale; NCCN, National Comprehensive Cancer Network; NR, not reported; SAOP2, Senior Adult Oncology Program 2; SPPB, Short Physical Performance Battery; TUG, Timed Up and Go test; TMTA, Trail Making Test A; TMTB, Trail Making Test B; WHOQOL, World Health Organization Quality of Life Questionnaire.

* ≥1 missing was also scored as abnormal CGA